

Product Carbon Footprint Study Report

Product Name: sanitary ceramics

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: sanitary ceramics

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg sanitary ceramics

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg sanitary ceramics.

2. System Boundary

sanitary ceramics production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of sanitary ceramics produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.761 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

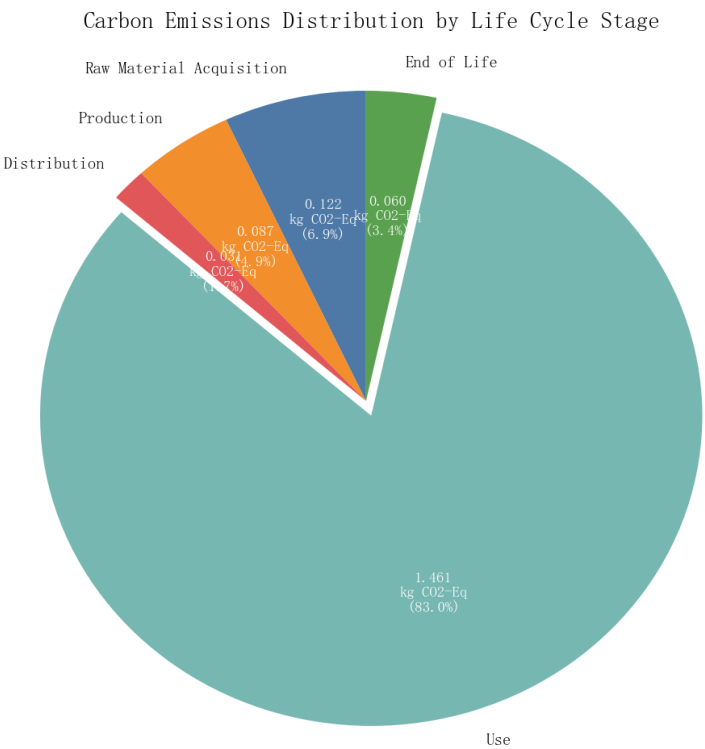


Table 1: Carbon Emissions by Life Cycle Stage

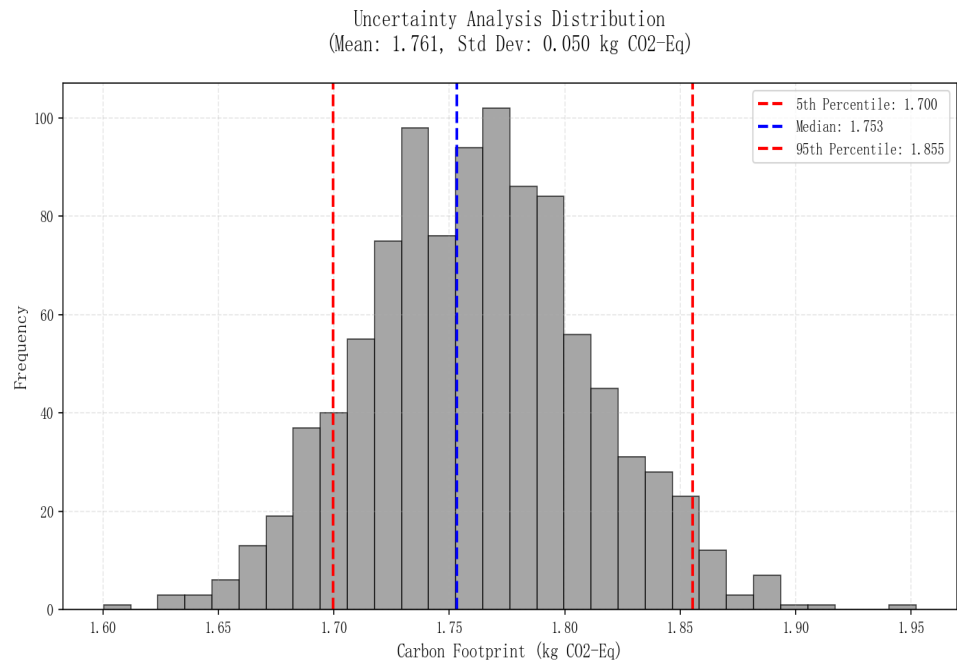
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.122	6.95
Production	0.087	4.93
Distribution	0.031	1.74

Use	1.461	82.96
End of Life	0.060	3.42
Total	1.761	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	1.761
Standard Deviation	0.050
Coefficient of Variation (CV)	2.82%
Median	1.753
5th Percentile	1.700
95th Percentile	1.855

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the sanitary ceramic product is 1.761 kg CO₂-Eq, which is relatively low compared to typical household appliances or construction materials, but meaningful when scaled across multiple units and lifetime use. The lifecycle stage distribution reveals a heavily use-dominated profile: the use phase contributes 1.461 kg CO₂-Eq (83.0%), underscoring that the environmental impact is overwhelmingly driven by operational energy or water heating rather than manufacturing, transport, or disposal. Raw material extraction (0.122 kg CO₂-Eq, 6.9%) and production (0.087 kg CO₂-Eq, 4.9%) together account for only about 12% of the total, while distribution (1.7%) and end-of-life (3.4%) are minor. These figures indicate that reducing the carbon footprint of sanitary ceramics requires a clear strategic focus on the use phase. However, the absence of identified top processes (all listed as “Unknown” with zero contribution) suggests that the underlying process-level data are incomplete or aggregated, limiting the ability to pinpoint specific emission drivers within each stage.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a fixed product lifespan, standardized use patterns (e.g., frequency of flushing, water temperature, or electricity consumption), and constant emission factors for energy and materials. The use phase emissions likely assume a specific energy mix (e.g., grid electricity for water heating) and a defined functional unit (e.g., one piece of sanitary ware over its expected lifetime). Data limitations are evident: the “Top Process Contributions” are entirely unknown, indicating that the inventory relies on generic or proxy datasets rather than facility-specific measurements. This introduces uncertainty in allocating emissions to individual processes such as clay mining, kiln firing, or packaging. The coefficient of variation (2.82%) is low, suggesting good statistical precision in the total estimate, but this precision may mask systematic biases from incomplete data. System boundary restrictions include a cradle-to-grave scope, but likely exclude capital goods (e.g., factory construction), maintenance, and any secondary product uses. The end-of-life stage may only consider landfill or incineration without recycling credits.

3. Suggestions for Improvement

- ****Target use-phase efficiency****: Implement water-saving technologies (e.g., dual-flush systems, low-flow designs) and promote cold-water connections to reduce energy for water heating. Provide clear labeling to encourage users to choose efficient models, potentially cutting the 83% use-phase contribution by 20–30%.
- ****Optimize raw material and production****: Substitute high-emission clays with recycled or locally sourced materials to reduce transport and extraction impacts. Improve kiln energy efficiency (e.g., switch to natural gas or electrification with renewables) to lower the production stage's 4.9% share.
- ****Enhance distribution logistics****: Consolidate shipments and use rail or electric trucks for last-mile delivery to reduce the already low 1.7% distribution impact, while also lowering other environmental burdens.

- ****Improve data quality****: Collect primary data from suppliers for the top emitting processes—particularly for raw material processing and firing—to replace “Unknown” contributions and reduce uncertainty. Conduct a full bill-of-materials review to identify missing process components.
- ****End-of-life valorization****: Design ceramics for easy disassembly and recyclability (e.g., as aggregate in concrete) to reduce the 3.4% end-of-life impact and create circularity benefits beyond the current carbon footprint.